CODEFORCES
Sponsored by TON

aryan_decoded06 | Logout

HOME TOP CATALOG CONTESTS GYM PROBLEMSET GROUPS RATING EDU API CALENDAR HELP

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS HACKS ROOM STANDINGS CUSTOM INVOCATION

C. Grid Covering

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Prakul has been working hard setting problems for Codecraft. When he's deep in thought, he likes jumping around his room in weird but specific ways. After some time, he wonders if jumping in this manner can cover all the tiles of his room.

Prakul's room can be considered as a grid consisting of rows and columns. He starts walking from . If he is currently at he can do either of the following moves:

- Jump steps right to , or
- Jump steps down to .

A special restriction is that he can start with either move, but **must alternate** between them.

Note that his room is quite special as it wraps around itself. Moving one step right from the -th column places him in the -st column. Similarly, moving one step down from the -th row places him in the -st row.

Since he still has to work on problemsetting, he needs your help. Determine if he can visit all tiles in in a finite number of jumps.

Input

Each test contains multiple test cases. The first line contains the number of test cases ( ). The description of the test cases follows.

The only line of each test case contains four integers ( ).

Output

For each test case, print "YES" if Prakul can cover all the tiles of his room, and "NO" otherwise.

You can output the answer in any case (upper or lower). For example, the strings "yEs", "yes", "Yes", and "YES" will be recognized as positive responses.

Example

input Copy

```
10
1 1 1 1
2 2 1 1
4 2 2 1
6 9 6 7
67 42 42 67
3411 4134 32 23
90234 143124 232 323
69387963 98793214 9791 4324786
985865 578977 899368 447605
1000000000 1000000000 1000000000 1000000000
```

output Copy

```
YES
YES
NO
NO
YES
NO
```

Codeforces Round 1091 (Div. 2) and CodeCraft 26
Finished
Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
370151523	Apr/07/2026 18:20	Accepted

→ Problem tags

chinese remainder theorem math number theory

No tag edit access

→ Contest materials

- Announcement (en)
- Tutorial (en)

```
NO
NO
YES
NO
```

Note

In the second test case: , , , . One possible way to cover the grid is if Prakul starts by moving **down**:

- Start at .
- Jump down tile: .
- Jump right tile: .
- Jump down tile: .
- Jump right tile: .

The sequence of visited tiles is . In this case, all tiles are covered, so the answer would be "YES". *Notice that Prakul wraps around the grid in the last two moves.*

In the third test case: , , , . If Prakul starts by moving **down**, the moves look like:

- Start at .
- Jump down tiles: .
- Jump right tile: .
- Jump down tiles: .
- Jump right tile: .
-

In this case, the tiles remain unvisited forever. Since not all tiles can be covered, the answer is "NO". It can be shown that even if Prakul started by moving **right** in the first move, the answer would not have changed.

Codeforces (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Apr/12/2026 04:58:46 UTC+5.5 (g1).
Desktop version, switch to mobile version.
Privacy Policy | Terms and Conditions

Supported by